

Master Thesis Summary

Comparative Study on the Influence of Electromagnetic Excitation Profiles on Electric Drivetrain Noise Based on Numerically and Experimentally Derived Forces

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1. Motivation and Problem Statement

With the rapid electrification of passenger and commercial powertrains, the acoustic signature of vehicles is no longer masked by internal combustion engines. In permanent magnet synchronous motors (PMSMs), high-frequency tonal noise, commonly known as acoustic whine, is directly generated by electromagnetic air gap forces exciting the structural resonances of the stator, rotor, and motor housing. While numerical electromagnetic simulations provide detailed air gap force predictions, they are computationally intensive and sensitive to material non-linearities and manufacturing tolerances. Conversely, experimental force measurement techniques, such as flex-PCB coil sensors placed in the air gap, offer real-world ground truth but require complex signal processing and spatial decomposition. This thesis investigates the quantitative relationship between electromagnetic force profiles and structural vibration, aiming to determine whether numerically derived or experimentally measured forces more accurately reproduce observed NVH behavior and to pinpoint the specific force harmonics that govern drivetrain noise.

2. Methodological Approach and Implementation

- **High-Fidelity FE Model Development in ANSYS:** Developed a complete three-dimensional structural finite element model of the motor from the ground up in ANSYS Mechanical, including the laminated stator core, copper windings, rotor assembly, housing, and bearing supports. Validated the structural model against an established NASTRAN reference model validated by Valeo and correlated mode shapes against experimental modal analysis (EMA).
- **Spatial and Temporal Force Decomposition:** Extracted electromagnetic excitation forces from both numerical electromagnetic FEA and experimental flex-PCB sensor measurements across varying operational torque-speed conditions. Performed two-dimensional Fourier transformation to decompose air gap forces into spatial orders (r) and temporal frequency harmonics (f).
- **Harmonic Response and Sensitivity Simulations:** Formulated harmonic response simulations applying reconstructed radial and tangential forces to stator teeth and rotor surfaces. Evaluated housing vibration velocities and Equivalent Radiated Power (ERP) levels. Conducted systematic sensitivity analyses to isolate the individual contributions of radial versus tangential force components across critical acoustic frequency bands.

3. Key Results and Scientific Insights

- **Dominance of Tangential Forces on Housing ERP:** Sensitivity analysis revealed that housing vibration and Equivalent Radiated Power are predominantly governed by tangential force harmonics rather than radial forces. Specifically, tangential force spatial mode $r = 0$ was identified as the primary contributor to acoustic excitation in the studied frequency range, challenging conventional assumptions that focus solely on radial deformation.
- **Simulation-to-Measurement Harmonic Discrepancies:** Detailed comparison of measured and simulated forces uncovered critical harmonic divergences. A significant excitation mode (force mode 1) was observed to exert substantial impact on structural response, highlighting gaps in nominal electromagnetic simulation models that fail to capture real-world rotor eccentricities and dynamic air gap asymmetries.
- **Multi-Physics Validation Paradigm:** Demonstrated that while structural mode shapes exhibited strong correlation between simulation and experiment, accurate acoustic noise prediction is exceptionally sensitive to precise electromagnetic force harmonic profiles. Reliable NVH design therefore necessitates validated multi-physical excitation inputs rather than purely structural refinement.

4. Core Competencies Demonstrated

Multi-Physics Modeling: Electromagnetic-structural coupling, modal extraction, harmonic response analysis, and Equivalent Radiated Power (ERP) evaluation.

Numerical & Experimental Tools: ANSYS Mechanical, NASTRAN benchmark validation, Hypermesh, flex-PCB sensor instrumentation, accelerometer data acquisition.

Scientific Computing & Signal Processing: Python, MATLAB, 2D Fourier transformation, spatial/temporal harmonic decomposition, and experimental modal correlation.